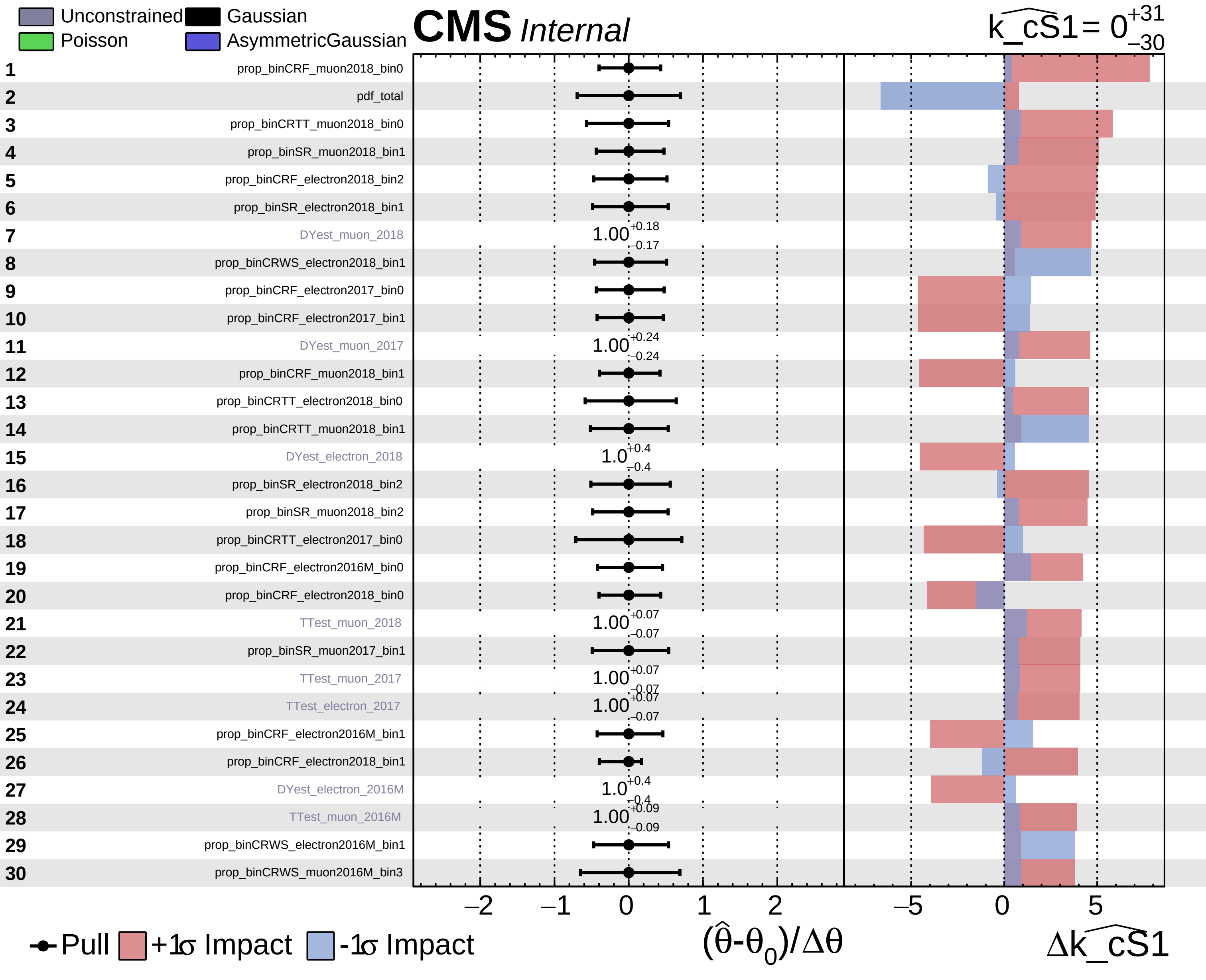
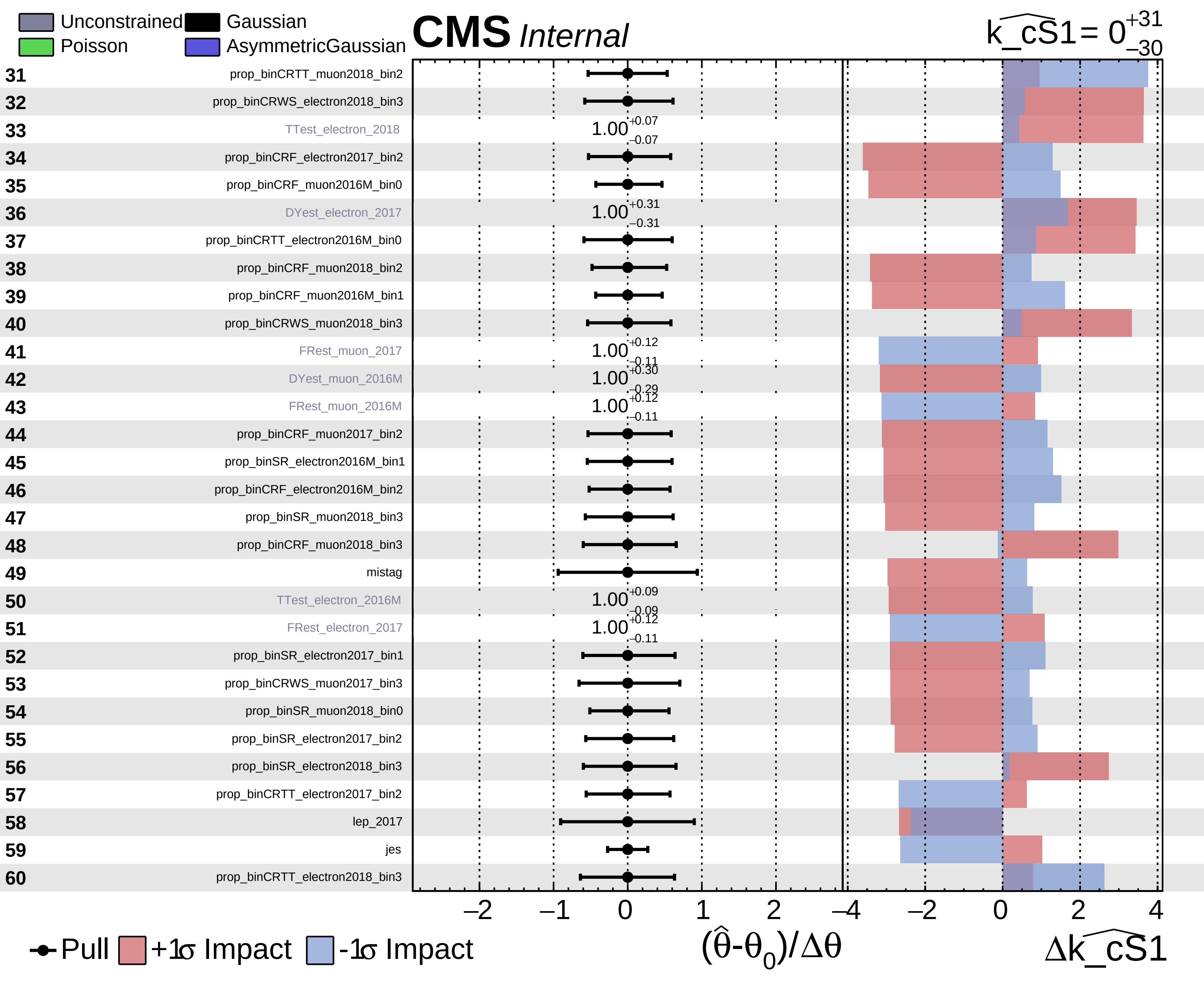
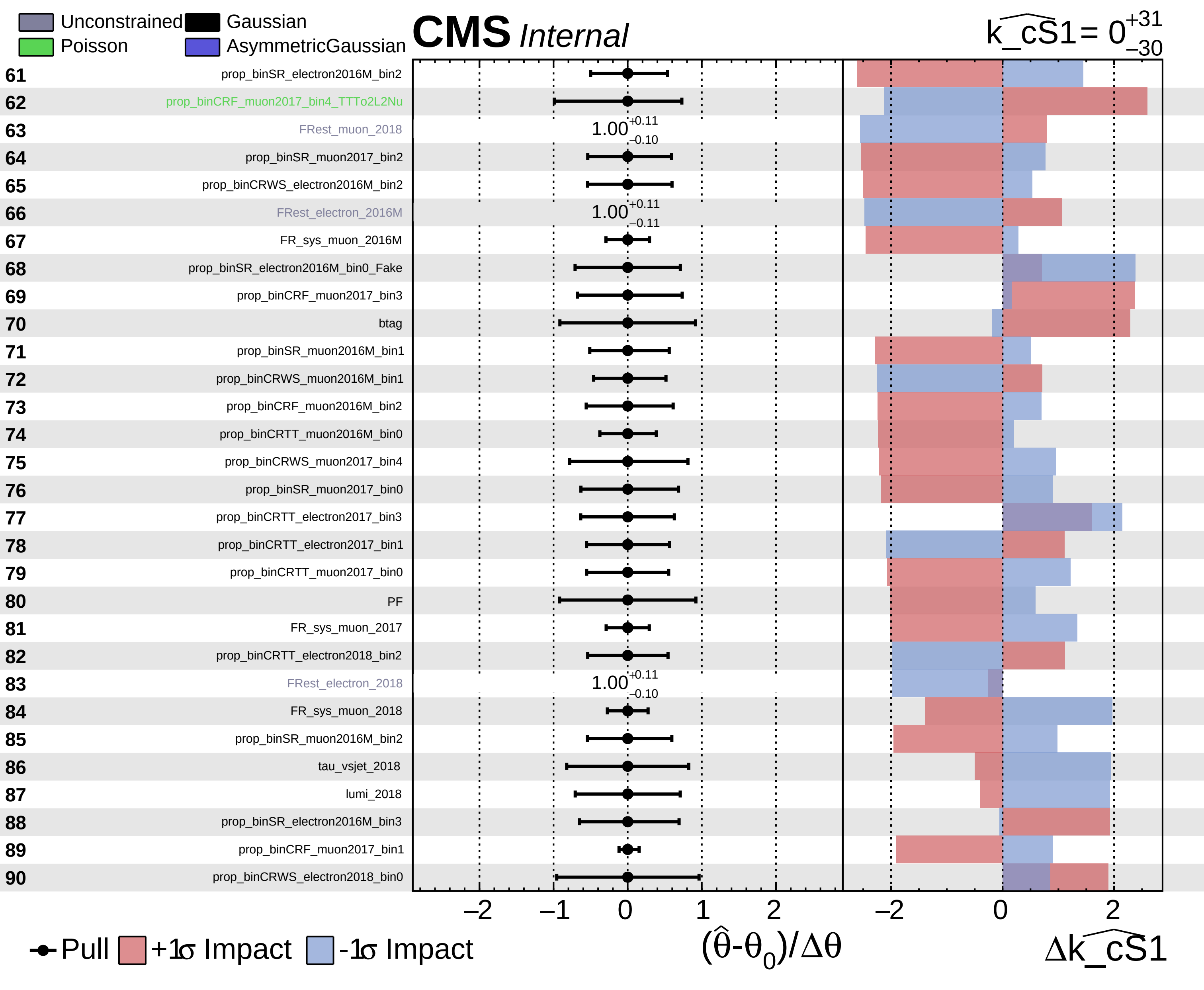


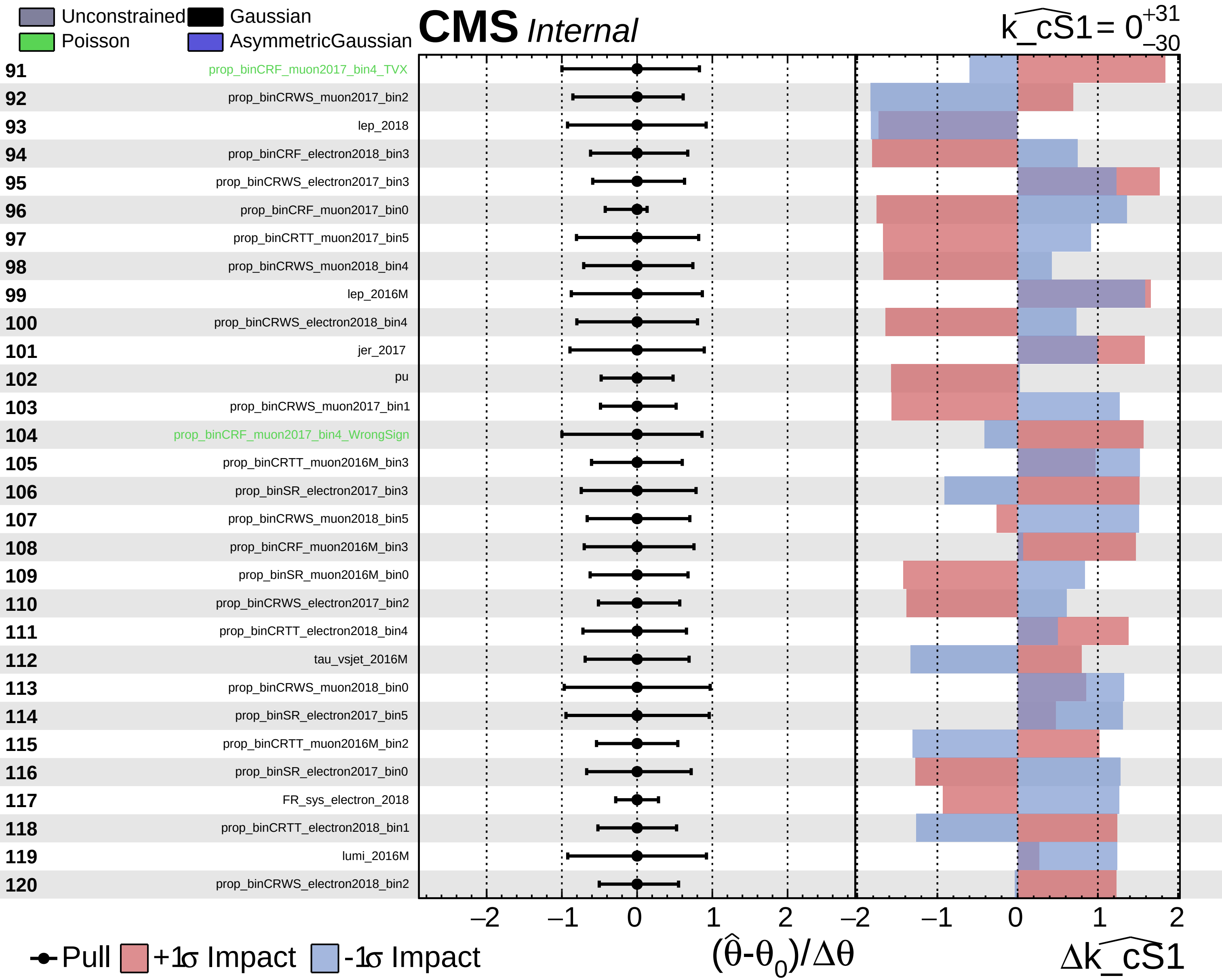
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cS1 = 0 +31 −30
1 prop_binCRF_muon2018_bin0
2 pdf_total
3 prop_binCRTT_muon2018_bin0
4 prop_binSR_muon2018_bin1
5 prop_binCRF_electron2018_bin2
6 prop_binSR_electron2018_bin1
7 DYest_muon_2018 1.00 +0.18 −0.17
8 prop_binCRWS_electron2018_bin1
9 prop_binCRF_electron2017_bin0
10 prop_binCRF_electron2017_bin1
11 DYest_muon_2017 1.00 +0.24 −0.24
12 prop_binCRF_muon2018_bin1
13 prop_binCRTT_electron2018_bin0
14 prop_binCRTT_muon2018_bin1
15 DYest_electron_2018 1.0 +0.4 −0.4
16 prop_binSR_electron2018_bin2
17 prop_binSR_muon2018_bin2
18 prop_binCRTT_electron2017_bin0
19 prop_binCRF_electron2016M_bin0
20 prop_binCRF_electron2018_bin0
21 TTest_muon_2018 1.00 +0.07 −0.07
22 prop_binSR_muon2017_bin1
23 TTest_muon_2017 1.00 +0.07 −0.07
24 TTest_electron_2017 1.00 +0.07 −0.07
25 prop_binCRF_electron2016M_bin1
26 prop_binCRF_electron2018_bin1
27 DYest_electron_2016M 1.0 +0.4 −0.4
28 TTest_muon_2016M 1.00 +0.09 −0.09
29 prop_binCRWS_electron2016M_bin1
30 prop_binCRWS_muon2016M_bin3
−2 −1 0 1 2
−5 0 5
Pull +1σ Impact -1σ Impact
(θ̂-θ0)/Δθ
Δk_cS1



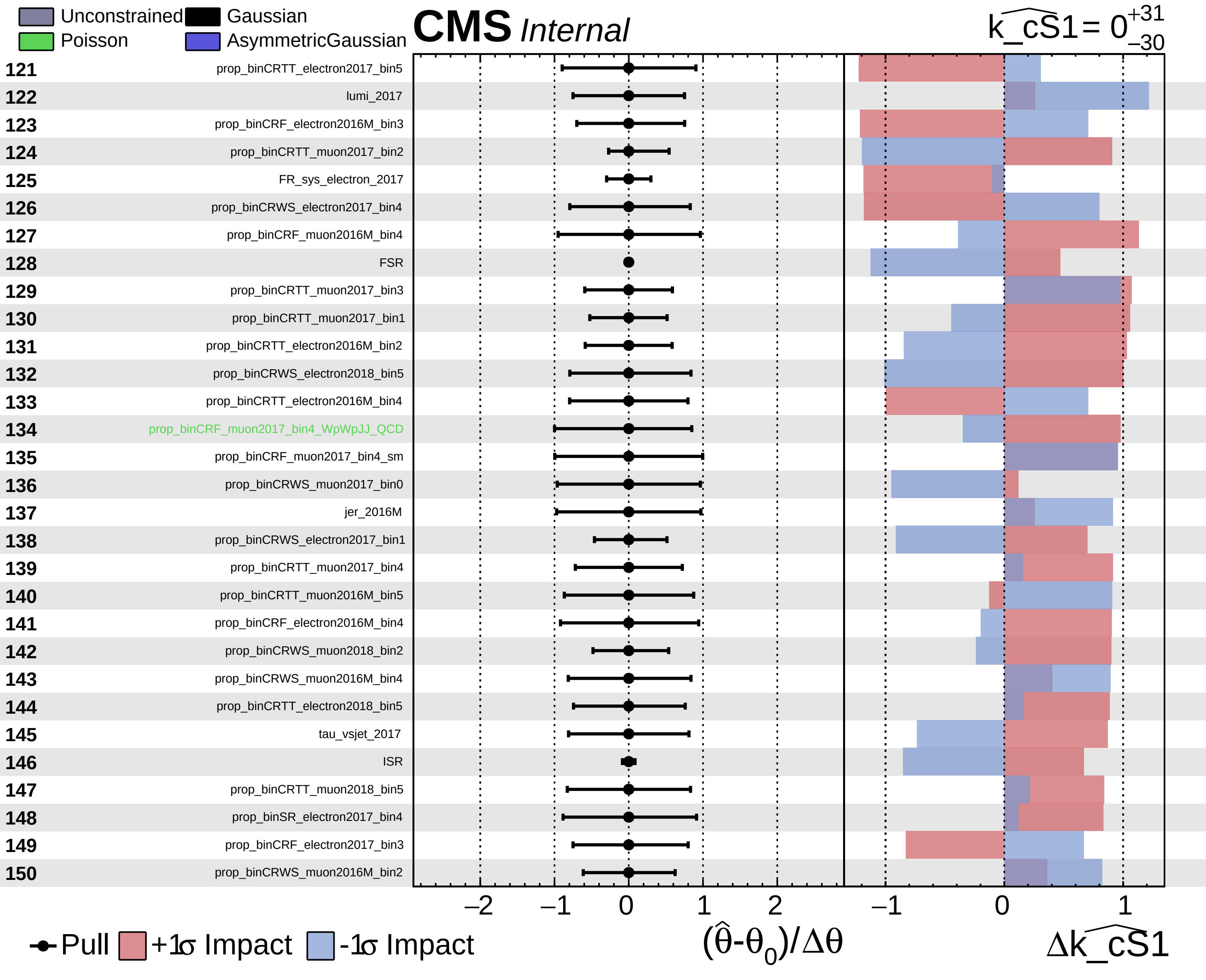
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cS1} = 0^{+31}_{-30}$
31 prop_binCRTT_muon2018_bin2
32 prop_binCRWS_electron2018_bin3
33 TTest_electron_2018 $1.00^{+0.07}_{-0.07}$
34 prop_binCRF_electron2017_bin2
35 prop_binCRF_muon2016M_bin0
36 DYest_electron_2017 $1.00^{+0.31}_{-0.31}$
37 prop_binCRTT_electron2016M_bin0
38 prop_binCRF_muon2018_bin2
39 prop_binCRF_muon2016M_bin1
40 prop_binCRWS_muon2018_bin3
41 FRest_muon_2017 $1.00^{+0.12}_{-0.11}$
42 DYest_muon_2016M $1.00^{+0.30}_{-0.29}$
43 FRest_muon_2016M $1.00^{+0.12}_{-0.11}$
44 prop_binCRF_muon2017_bin2
45 prop_binSR_electron2016M_bin1
46 prop_binCRF_electron2016M_bin2
47 prop_binSR_muon2018_bin3
48 prop_binCRF_muon2018_bin3
49 mistag
50 TTest_electron_2016M $1.00^{+0.09}_{-0.09}$
51 FRest_electron_2017 $1.00^{+0.12}_{-0.11}$
52 prop_binSR_electron2017_bin1
53 prop_binCRWS_muon2017_bin3
54 prop_binSR_muon2018_bin0
55 prop_binSR_electron2017_bin2
56 prop_binSR_electron2018_bin3
57 prop_binCRTT_electron2017_bin2
58 lep_2017
59 jes
60 prop_binCRTT_electron2018_bin3
−2 −1 0 1 2
−4 −2 0 2 4
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cS1}$



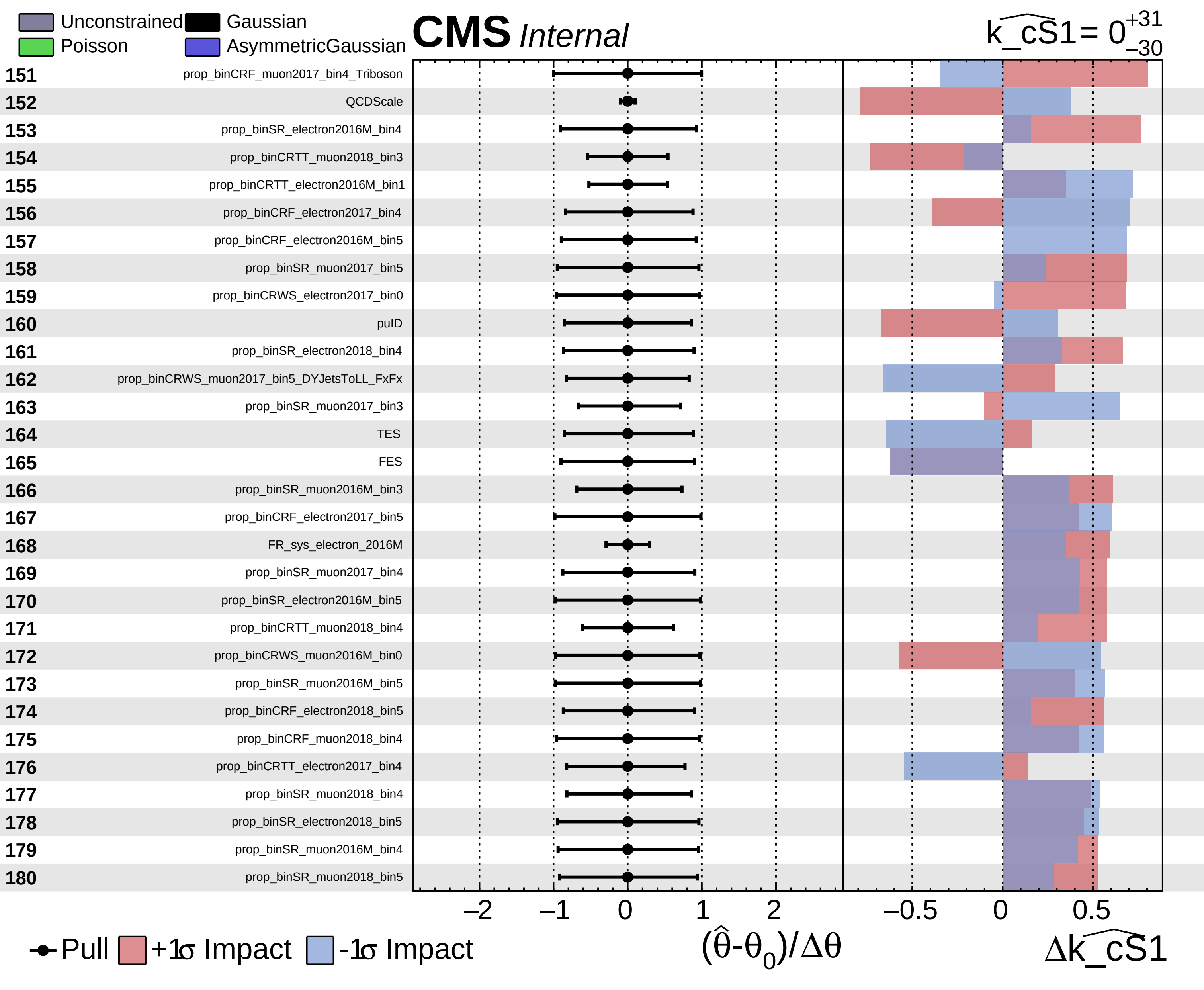
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cS1 = 0 +31 −30
61 prop_binSR_electron2016M_bin2
62 prop_binCRF_muon2017_bin4_TTTo2L2Nu
63 FRest_muon_2018 1.00 +0.11 −0.10
64 prop_binSR_muon2017_bin2
65 prop_binCRWS_electron2016M_bin2
66 FRest_electron_2016M 1.00 +0.11 −0.11
67 FR_sys_muon_2016M
68 prop_binSR_electron2016M_bin0_Fake
69 prop_binCRF_muon2017_bin3
70 btag
71 prop_binSR_muon2016M_bin1
72 prop_binCRWS_muon2016M_bin1
73 prop_binCRF_muon2016M_bin2
74 prop_binCRTT_muon2016M_bin0
75 prop_binCRWS_muon2017_bin4
76 prop_binSR_muon2017_bin0
77 prop_binCRTT_electron2017_bin3
78 prop_binCRTT_electron2017_bin1
79 prop_binCRTT_muon2017_bin0
80 PF
81 FR_sys_muon_2017
82 prop_binCRTT_electron2018_bin2
83 FRest_electron_2018 1.00 +0.11 −0.10
84 FR_sys_muon_2018
85 prop_binSR_muon2016M_bin2
86 tau_vsjet_2018
87 lumi_2018
88 prop_binSR_electron2016M_bin3
89 prop_binCRF_muon2017_bin1
90 prop_binCRWS_electron2018_bin0
−2 −1 0 1 2
−2 0 2
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cS1



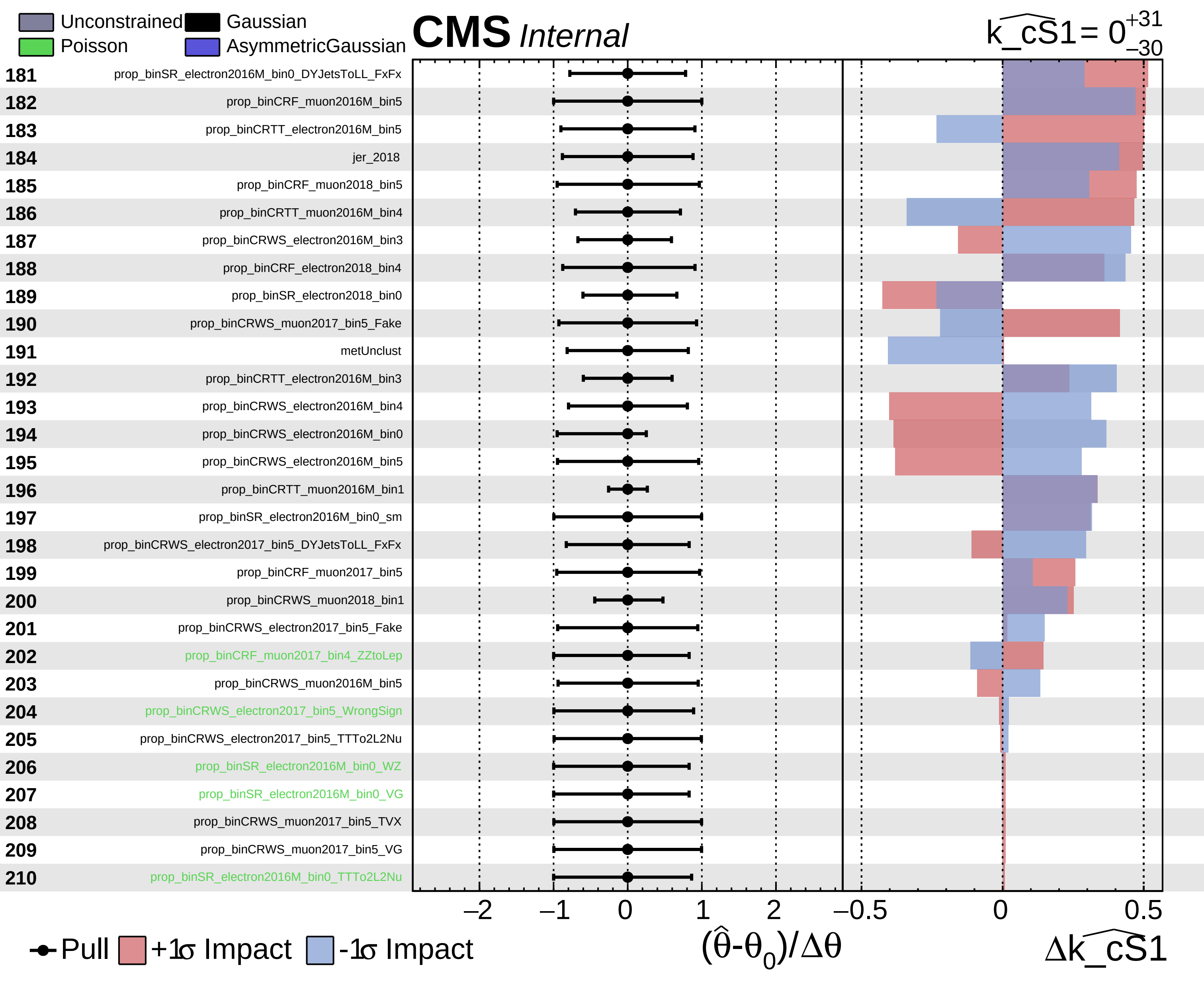
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cS1} = 0^{+31}_{-30}$
91 prop_binCRF_muon2017_bin4_TVX
92 prop_binCRWS_muon2017_bin2
93 lep_2018
94 prop_binCRF_electron2018_bin3
95 prop_binCRWS_electron2017_bin3
96 prop_binCRF_muon2017_bin0
97 prop_binCRTT_muon2017_bin5
98 prop_binCRWS_muon2018_bin4
99 lep_2016M
100 prop_binCRWS_electron2018_bin4
101 jer_2017
102 pu
103 prop_binCRWS_muon2017_bin1
104 prop_binCRF_muon2017_bin4_WrongSign
105 prop_binCRTT_muon2016M_bin3
106 prop_binSR_electron2017_bin3
107 prop_binCRWS_muon2018_bin5
108 prop_binCRF_muon2016M_bin3
109 prop_binSR_muon2016M_bin0
110 prop_binCRWS_electron2017_bin2
111 prop_binCRTT_electron2018_bin4
112 tau_vsjet_2016M
113 prop_binCRWS_muon2018_bin0
114 prop_binSR_electron2017_bin5
115 prop_binCRTT_muon2016M_bin2
116 prop_binSR_electron2017_bin0
117 FR_sys_electron_2018
118 prop_binCRTT_electron2018_bin1
119 lumi_2016M
120 prop_binCRWS_electron2018_bin2
−2 −1 0 1 2
−2 −1 0 1 2
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cS1}$



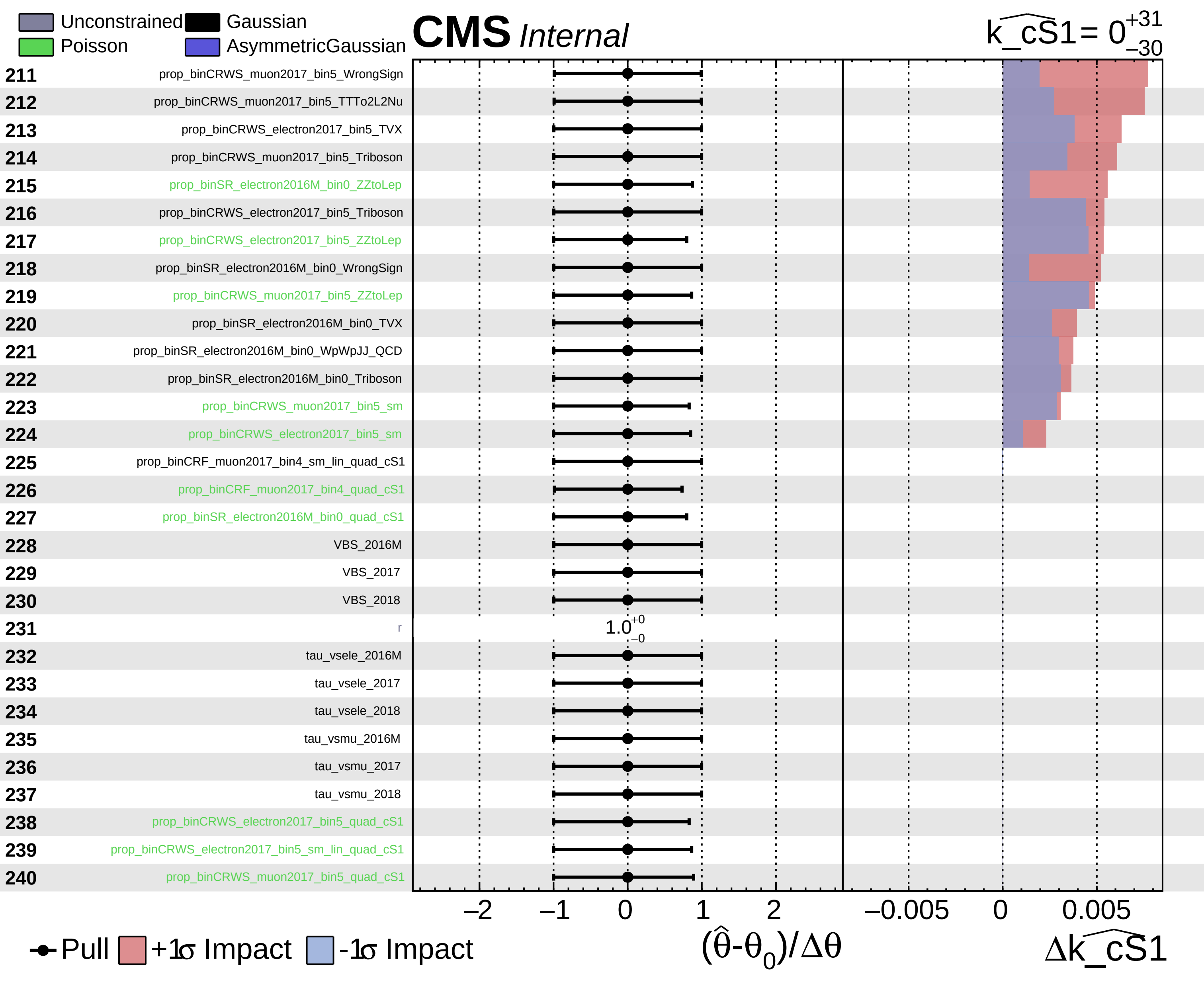
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cS1} = 0^{+31}_{-30}$
121 prop_binCRTT_electron2017_bin5
122 lumi_2017
123 prop_binCRF_electron2016M_bin3
124 prop_binCRTT_muon2017_bin2
125 FR_sys_electron_2017
126 prop_binCRWS_electron2017_bin4
127 prop_binCRF_muon2016M_bin4
128 FSR
129 prop_binCRTT_muon2017_bin3
130 prop_binCRTT_muon2017_bin1
131 prop_binCRTT_electron2016M_bin2
132 prop_binCRWS_electron2018_bin5
133 prop_binCRTT_electron2016M_bin4
134 prop_binCRF_muon2017_bin4_WpWpJJ_QCD
135 prop_binCRF_muon2017_bin4_sm
136 prop_binCRWS_muon2017_bin0
137 jer_2016M
138 prop_binCRWS_electron2017_bin1
139 prop_binCRTT_muon2017_bin4
140 prop_binCRTT_muon2016M_bin5
141 prop_binCRF_electron2016M_bin4
142 prop_binCRWS_muon2018_bin2
143 prop_binCRWS_muon2016M_bin4
144 prop_binCRTT_electron2018_bin5
145 tau_vsjet_2017
146 ISR
147 prop_binCRTT_muon2018_bin5
148 prop_binSR_electron2017_bin4
149 prop_binCRF_electron2017_bin3
150 prop_binCRWS_muon2016M_bin2
−2 −1 0 1 2
−1 0 1
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cS1}$



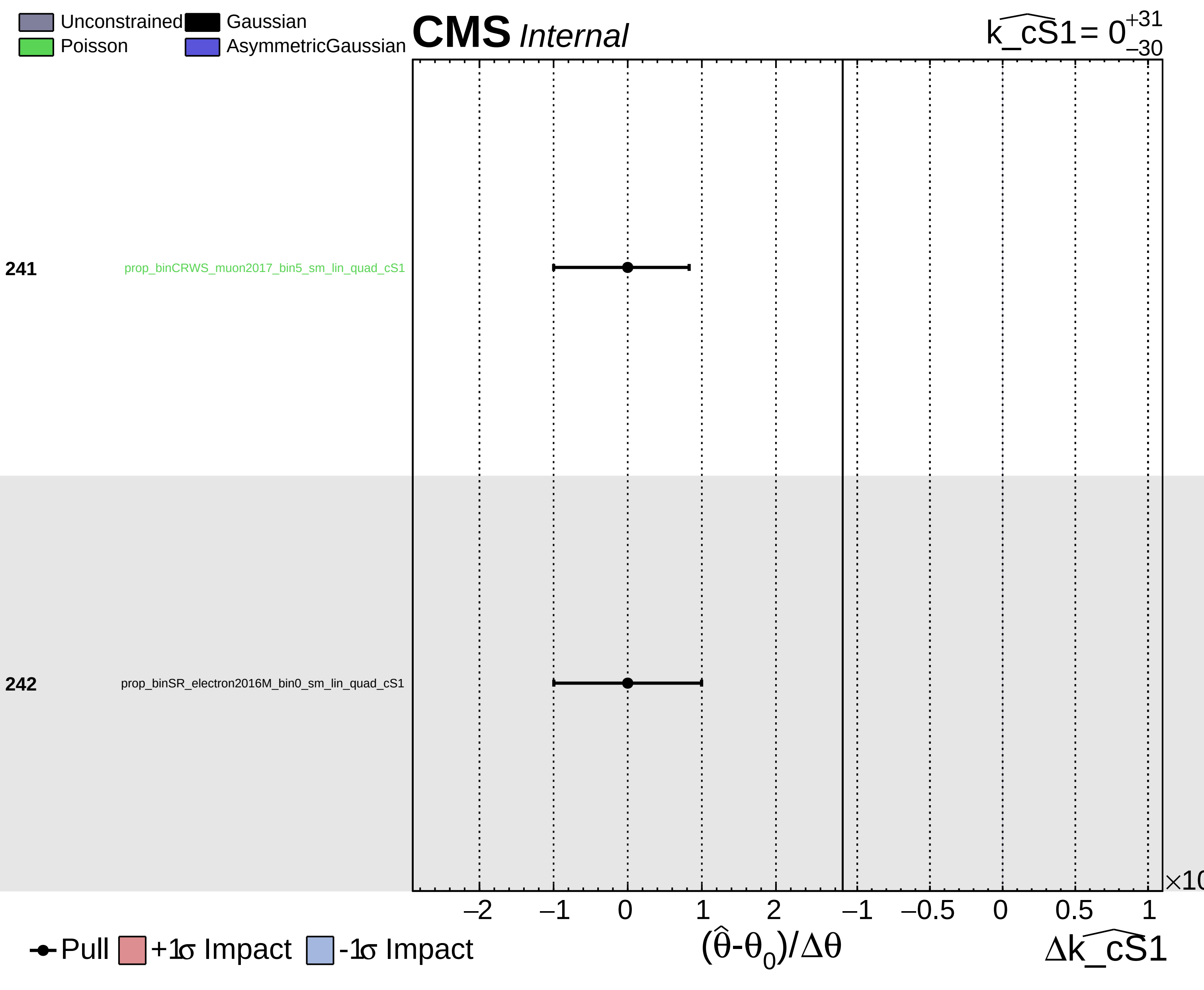
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cS1 = 0 +31 −30
151 prop_binCRF_muon2017_bin4_Triboson
152 QCDScale
153 prop_binSR_electron2016M_bin4
154 prop_binCRTT_muon2018_bin3
155 prop_binCRTT_electron2016M_bin1
156 prop_binCRF_electron2017_bin4
157 prop_binCRF_electron2016M_bin5
158 prop_binSR_muon2017_bin5
159 prop_binCRWS_electron2017_bin0
160 puID
161 prop_binSR_electron2018_bin4
162 prop_binCRWS_muon2017_bin5_DYJetsToLL_FxFx
163 prop_binSR_muon2017_bin3
164 TES
165 FES
166 prop_binSR_muon2016M_bin3
167 prop_binCRF_electron2017_bin5
168 FR_sys_electron_2016M
169 prop_binSR_muon2017_bin4
170 prop_binSR_electron2016M_bin5
171 prop_binCRTT_muon2018_bin4
172 prop_binCRWS_muon2016M_bin0
173 prop_binSR_muon2016M_bin5
174 prop_binCRF_electron2018_bin5
175 prop_binCRF_muon2018_bin4
176 prop_binCRTT_electron2017_bin4
177 prop_binSR_muon2018_bin4
178 prop_binSR_electron2018_bin5
179 prop_binSR_muon2016M_bin4
180 prop_binSR_muon2018_bin5
−2
−1
0
1
2
−0.5
0
0.5
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cS1



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cS1 = 0 +31 −30
181 prop_binSR_electron2016M_bin0_DYJetsToLL_FxFx
182 prop_binCRF_muon2016M_bin5
183 prop_binCRTT_electron2016M_bin5
184 jer_2018
185 prop_binCRF_muon2018_bin5
186 prop_binCRTT_muon2016M_bin4
187 prop_binCRWS_electron2016M_bin3
188 prop_binCRF_electron2018_bin4
189 prop_binSR_electron2018_bin0
190 prop_binCRWS_muon2017_bin5_Fake
191 metUnclust
192 prop_binCRTT_electron2016M_bin3
193 prop_binCRWS_electron2016M_bin4
194 prop_binCRWS_electron2016M_bin0
195 prop_binCRWS_electron2016M_bin5
196 prop_binCRTT_muon2016M_bin1
197 prop_binSR_electron2016M_bin0_sm
198 prop_binCRWS_electron2017_bin5_DYJetsToLL_FxFx
199 prop_binCRF_muon2017_bin5
200 prop_binCRWS_muon2018_bin1
201 prop_binCRWS_electron2017_bin5_Fake
202 prop_binCRF_muon2017_bin4_ZZtoLep
203 prop_binCRWS_muon2016M_bin5
204 prop_binCRWS_electron2017_bin5_WrongSign
205 prop_binCRWS_electron2017_bin5_TTTo2L2Nu
206 prop_binSR_electron2016M_bin0_WZ
207 prop_binSR_electron2016M_bin0_VG
208 prop_binCRWS_muon2017_bin5_TVX
209 prop_binCRWS_muon2017_bin5_VG
210 prop_binSR_electron2016M_bin0_TTTo2L2Nu
−2 −1 0 1 2
−0.5 0 0.5
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cS1}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cS1 = 0+31−30
211 prop_binCRWS_muon2017_bin5_WrongSign
212 prop_binCRWS_muon2017_bin5_TTTo2L2Nu
213 prop_binCRWS_electron2017_bin5_TVX
214 prop_binCRWS_muon2017_bin5_Triboson
215 prop_binSR_electron2016M_bin0_ZZtoLep
216 prop_binCRWS_electron2017_bin5_Triboson
217 prop_binCRWS_electron2017_bin5_ZZtoLep
218 prop_binSR_electron2016M_bin0_WrongSign
219 prop_binCRWS_muon2017_bin5_ZZtoLep
220 prop_binSR_electron2016M_bin0_TVX
221 prop_binSR_electron2016M_bin0_WpWpJJ_QCD
222 prop_binSR_electron2016M_bin0_Triboson
223 prop_binCRWS_muon2017_bin5_sm
224 prop_binCRWS_electron2017_bin5_sm
225 prop_binCRF_muon2017_bin4_sm_lin_quad_cS1
226 prop_binCRF_muon2017_bin4_quad_cS1
227 prop_binSR_electron2016M_bin0_quad_cS1
228 VBS_2016M
229 VBS_2017
230 VBS_2018
231 r
1.0+0−0
232 tau_vsele_2016M
233 tau_vsele_2017
234 tau_vsele_2018
235 tau_vsmu_2016M
236 tau_vsmu_2017
237 tau_vsmu_2018
238 prop_binCRWS_electron2017_bin5_quad_cS1
239 prop_binCRWS_electron2017_bin5_sm_lin_quad_cS1
240 prop_binCRWS_muon2017_bin5_quad_cS1
−2
−1
0
1
2
−0.005
0
0.005
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cS1



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cS1 = 0 +31 −30
241
prop_binCRWS_muon2017_bin5_sm_lin_quad_cS1
242
prop_binSR_electron2016M_bin0_sm_lin_quad_cS1
−2 −1 0 1 2
−1 −0.5 0 0.5 1
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cS1